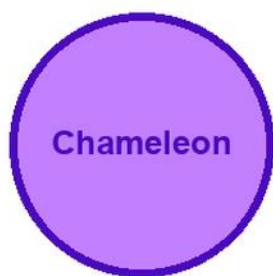


KITABU QUEST

P5

SCIENCE AND ELEMENTARY TECHNOLOGY



Cover scene

learners doing a simple science and technology investigation with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

5

Title Page

KITABU QUEST P5 Science and Elementary Technology

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P5 learners study Science and Elementary Technology one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Observe, ask questions, design safely, record evidence, and explain findings.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Carpentry tools
2. Masonry tools
3. Light
4. Computer memories and storages
5. Writing skills on computer
6. Computer research
7. Programming for children
8. Water pollution
9. Soil and cultivation
10. Plants and environment
11. Human digestive system
12. Human diseases

As you finish each topic, return to the success criteria and correct one answer before moving on.

Carpentry tools: Lesson Opener

Learning focus: Carpentry tools

Country link: a safe Rwandan observation, model, data table, or investigation for Carpentry tools.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- name different carpentry tools, explain how to use and care for them, use them safely in simple tasks, and show responsibility and interest in safe carpentry work
- identify common carpentry tools
- explain what each tool is used for
- describe how to take care of carpentry tools
- work in groups to differentiate various carpentry tools according to their various usages
- practice the activity of fixing and removing nails from the wood

Inquiry questions:

- How can carpentry tools help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carpentry tools: Learn

Carpentry tools is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Carpentry tools
- Carpentry
- tools
- concept
- observation
- evidence
- safety
- conclusion

Carpentry tools: Worked Example

Investigation model for Carpentry tools: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Carpentry tools: Vocabulary And Study Skill

Use these words and study moves before you practise Carpentry tools. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Carpentry tools
- Carpentry
- tools
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Carpentry tools without a clear example, method, or evidence.

Carpentry tools: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Carpentry tools.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Carpentry tools: Practice And Correction

Practice for Carpentry tools:

Main outcome: name different carpentry tools, explain how to use and care for them, use them safely in simple tasks, and show responsibility and interest in safe carpentry work.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Carpentry tools: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Carpentry tools.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carpentry tools: Outcome Check

Outcome check for Carpentry tools: Name different carpentry tools, explain how to use and care for them, use them safely in simple tasks, and show responsibility and interest in safe carpentry work.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carpentry tools: Outcome Check

Outcome check for Carpentry tools: Identify common carpentry tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Masonry tools: Lesson Opener

Learning focus: Masonry tools

Country link: a safe Rwandan observation, model, data table, or investigation for Masonry tools.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- name masonry tools, explain how to use and care for them, handle them safely during simple tasks, and show responsibility in using and maintaining masonry tools
- identify masonry tools
- explain the use of masonry tools
- describe dangers of using masonry tools
- explain how to avoid accidents when using masonry tools
- clean and store masonry tools properly

Inquiry questions:

- How can masonry tools help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Masonry tools: Learn

Masonry tools is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Masonry tools
- Masonry
- tools
- concept
- observation
- evidence
- safety
- conclusion

Masonry tools: Worked Example

Investigation model for Masonry tools: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Masonry tools: Vocabulary And Study Skill

Use these words and study moves before you practise Masonry tools. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Masonry tools
- Masonry
- tools
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Masonry tools without a clear example, method, or evidence.

Masonry tools: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Masonry tools.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Masonry tools: Practice And Correction

Practice for Masonry tools:

Main outcome: name masonry tools, explain how to use and care for them, handle them safely during simple tasks, and show responsibility in using and maintaining masonry tools.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Masonry tools: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Masonry tools.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Masonry tools: Outcome Check

Outcome check for Masonry tools: Name masonry tools, explain how to use and care for them, handle them safely during simple tasks, and show responsibility in using and maintaining masonry tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Masonry tools: Outcome Check

Outcome check for Masonry tools: Identify masonry tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light: Lesson Opener

Learning focus: Light

Country link: a safe Rwandan observation, model, data table, or investigation for Light.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- describe how light moves and behaves, perform simple experiments to observe how light travels, reflects, and bends, and show interest by exploring light in their surroundings
- discuss how light moves
- demonstrate how light passes through different materials
- identify different types of materials: transparent, translucent, and opaque
- watch and describe how light moves or travels
- do experiments to show light passing through materials

Inquiry questions:

- How can light help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light: Learn

Light explains how light travels and reflects.

Core idea:

- Light travels in straight lines through a uniform medium.
- Shadows form when an opaque object blocks light.
- Reflection occurs when light bounces from a surface.
- Smooth, shiny surfaces reflect light more regularly than rough surfaces.

Key words:

- Light
- concept
- observation
- evidence
- safety
- conclusion

Light: Worked Example

Investigation model for Light: Show that light travels in straight lines.

1. Make small holes in three cards.
2. Place a torch behind the first card.
3. Align the holes so light reaches a screen.
4. Move one card sideways and observe the screen.

Conclusion: the light path is blocked when the holes are not in a straight line.

Light: Vocabulary And Study Skill

Use these words and study moves before you practise Light. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Light
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Light without a clear example, method, or evidence.

Light: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Light.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Light: Practice And Correction

Practice for Light:

Main outcome: describe how light moves and behaves, perform simple experiments to observe how light travels, reflects, and bends, and show interest by exploring light in their surroundings.

1. Draw a straight ray of light from a torch.
2. Explain how a shadow forms.
3. Name one smooth reflector and one rough surface.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Light: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Light.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light: Outcome Check

Outcome check for Light: Describe how light moves and behaves, perform simple experiments to observe how light travels, reflects, and bends, and show interest by exploring light in their surroundings.

1. Draw a straight light ray.
2. Explain how a shadow forms.
3. Describe reflection from a smooth surface.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light: Outcome Check

Outcome check for Light: Discuss how light moves.

1. Draw a straight light ray.
2. Explain how a shadow forms.
3. Describe reflection from a smooth surface.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer memories and storages: Lesson Opener

Learning focus: Computer memories and storages

Country link: a safe Rwandan observation, model, data table, or investigation for Computer memories and storages.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- identify different types of computer memory and storage and their use in storing information both offline and online
- explain the concept of data, memory and storage and data sharing
- identify the different computer memories and storages found inside and outside the computer
- list the different memory units and their relationships in terms of size
- use different types of storage and memory devices
- type, save and open a document from internal storage

Inquiry questions:

- How can computer memories and storages help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer memories and storages: Learn

Computer memories and storages teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Computer memories and storages
- Computer
- memories
- storages
- concept
- observation
- evidence
- safety

Computer memories and storages: Worked Example

Worked example for Computer memories and storages: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Computer memories and storages: Vocabulary And Study Skill

Use these words and study moves before you practise Computer memories and storages. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Computer memories and storages
- Computer
- memories
- storages
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Computer memories and storages without a clear example, method, or evidence.

Computer memories and storages: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Computer memories and storages.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Computer memories and storages: Practice And Correction

Practice for Computer memories and storages:

Main outcome: identify different types of computer memory and storage and their use in storing information both offline and online.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Computer memories and storages: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Computer memories and storages.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer memories and storages: Outcome Check

Outcome check for Computer memories and storages: Identify different types of computer memory and storage and their use in storing information both offline and online.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer memories and storages: Outcome Check

Outcome check for Computer memories and storages: Explain the concept of data, memory and storage and data sharing.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Writing skills on computer: Lesson Opener

Learning focus: Writing skills on computer

Country link: a safe Rwandan observation, model, data table, or investigation for Writing skills on computer.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- use a Word processing program to create and edit tables, insert and arrange pictures in a document
- distinguish between columns and rows of a table
- identify how to extend the length of a column and the height of row, and increase/ decrease the number of rows or columns
- create a table with a specific number of rows and columns
- add and edit data into a table
- edit the table by adding or deleting columns and rows

Inquiry questions:

- How can writing skills on computer help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Writing skills on computer: Learn

Writing skills on computer teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Writing skills on computer
- Writing
- skills
- computer
- concept
- observation
- evidence
- safety

Writing skills on computer: Worked Example

Worked example for Writing skills on computer: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Writing skills on computer: Vocabulary And Study Skill

Use these words and study moves before you practise Writing skills on computer. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Writing skills on computer
- Writing
- skills
- computer
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Writing skills on computer without a clear example, method, or evidence.

Writing skills on computer: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Writing skills on computer.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Writing skills on computer: Practice And Correction

Practice for Writing skills on computer:

Main outcome: use a Word processing program to create and edit tables, insert and arrange pictures in a document.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Writing skills on computer: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Writing skills on computer.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Writing skills on computer: Outcome Check

Outcome check for Writing skills on computer: Use a Word processing program to create and edit tables, insert and arrange pictures in a document.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Writing skills on computer: Outcome Check

Outcome check for Writing skills on computer: Distinguish between columns and rows of a table.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer research: Lesson Opener

Learning focus: Computer research

Country link: a safe Rwandan observation, model, data table, or investigation for Computer research.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- perform online research using a computer and a computer equipped with AI tools
- define common internet terms
- describe what online research is and list examples of search engines
- identify learning resources available online
- explain internet safety practices
- use a search engine to locate accurate and relevant information online

Inquiry questions:

- How can computer research help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer research: Learn

Computer research teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Computer research
- Computer
- research
- concept
- observation
- evidence
- safety
- conclusion

Computer research: Worked Example

Worked example for Computer research: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Computer research: Vocabulary And Study Skill

Use these words and study moves before you practise Computer research. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Computer research
- Computer
- research
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Computer research without a clear example, method, or evidence.

Computer research: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Computer research.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Computer research: Practice And Correction

Practice for Computer research:

Main outcome: perform online research using a computer and a computer equipped with AI tools.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Computer research: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Computer research.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer research: Outcome Check

Outcome check for Computer research: Perform online research using a computer and a computer equipped with AI tools.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer research: Outcome Check

Outcome check for Computer research: Define common internet terms.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Lesson Opener

Learning focus: Programming for children

Country link: a safe Rwandan observation, model, data table, or investigation for Programming for children.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- draw geometric shapes, develop codes that perform arithmetic operations using turtle art program, create dialogue & cartoons using Scratch programs and create a program that give basic commands to a robot using Pictoblox
- explain how Turtle Art can be used to draw geometric shapes
- associate Turtle Art instructions with arithmetic operations to perform calculations
- associate different commands to make Scratch projects
- design and apply a new background for a Scratch project
- choose and add sound blocks to a Scratch program

Inquiry questions:

- How can programming for children help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Learn

Programming for children is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Programming for children: Worked Example

Worked example for Programming for children: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Programming for children: Vocabulary And Study Skill

Use these words and study moves before you practise Programming for children. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Programming for children without a clear example, method, or evidence.

Programming for children: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Programming for children.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Programming for children: Practice And Correction

Practice for Programming for children:

Main outcome: draw geometric shapes, develop codes that perform arithmetic operations using turtle art program, create dialogue & cartoons using Scratch programs and create a program that give basic commands to a robot using Pictoblox.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Programming for children: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Programming for children.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Draw geometric shapes, develop codes that perform arithmetic operations using turtle art program, create dialogue & cartoons using Scratch programs and create a program that give basic commands to a robot using Pictoblox.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Explain how Turtle Art can be used to draw geometric shapes.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Lesson Opener

Learning focus: Water pollution

Country link: a safe Rwandan observation, model, data table, or investigation for Water pollution.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain the importance of water, show how to keep water clean and safe, practice ways to protect the environment from rainwater damage, and show care for water sources in the community
- explain the good and bad effects of rainwater
- identify the dangers of rain water and how to prevent them
- explain methods of protecting the environment from rain water
- investigate and explain water pollution using evidence
- clean water by using filtration, boiling, and treatment tablets

Inquiry questions:

- How can water pollution help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Learn

Water pollution is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Water pollution
- Water
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Water pollution: Worked Example

Investigation model for Water pollution: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Water pollution: Vocabulary And Study Skill

Use these words and study moves before you practise Water pollution. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Water pollution
- Water
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Water pollution without a clear example, method, or evidence.

Water pollution: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Water pollution.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Water pollution: Practice And Correction

Practice for Water pollution:

Main outcome: explain the importance of water, show how to keep water clean and safe, practice ways to protect the environment from rainwater damage, and show care for water sources in the community.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Water pollution: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Water pollution.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Outcome Check

Outcome check for Water pollution: Explain the importance of water, show how to keep water clean and safe, practice ways to protect the environment from rainwater damage, and show care for water sources in the community.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Outcome Check

Outcome check for Water pollution: Explain the good and bad effects of rainwater.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil and cultivation: Lesson Opener

Learning focus: Soil and cultivation

Country link: a safe Rwandan observation, model, data table, or investigation for Soil and cultivation.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain how to prepare soil and use fertilizers properly, practice soil preparation and fertilizer use, and show respect and care for fertilizers to protect crops and the environment
- identify steps of soil preparation for cultivation
- explain the importance of fertilizers
- identify rules of applying fertilizers
- prepare soil ready for planting
- choose the right fertilizers

Inquiry questions:

- How can soil and cultivation help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil and cultivation: Learn

Soil and cultivation explains soil as a living natural resource that supports plants, animals, and people.

Core idea:

- Soil is made from weathered rock, humus, air, water, and living organisms.
- Sandy, clay, and loam soils feel different, hold water differently, and suit different uses.
- Healthy soil supports plant roots, stores water, and provides nutrients.
- Soil can be damaged by erosion, burning, pollution, and careless farming.
- Learners can compare soil samples safely by colour, texture, water retention, and visible organic matter.

Key words:

- Soil and cultivation
- Soil
- cultivation
- concept
- observation
- evidence
- safety
- conclusion

Soil and cultivation: Worked Example

Investigation model for Soil and cultivation: Compare two soil samples safely.

1. Collect two small teacher-approved soil samples, such as garden soil and sandy soil.
2. Observe colour and texture without putting soil near the mouth or eyes.
3. Add a small amount of water and compare how each sample holds water.
4. Record results in a table: sample, colour, texture, water retention, visible organic matter.
5. Explain which sample may support plant growth better and why.

Conclusion: soil properties affect plant growth and land use.

Soil and cultivation: Vocabulary And Study Skill

Use these words and study moves before you practise Soil and cultivation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Soil and cultivation
- Soil
- cultivation
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Soil and cultivation without a clear example, method, or evidence.

Soil and cultivation: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Soil and cultivation.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Soil and cultivation: Practice And Correction

Practice for Soil and cultivation:

Main outcome: explain how to prepare soil and use fertilizers properly, practice soil preparation and fertilizer use, and show respect and care for fertilizers to protect crops and the environment.

1. Compare two safe soil samples by colour, texture, and water retention.
2. Record observations in a table.
3. Explain which soil may support plant growth better and why.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Soil and cultivation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Soil and cultivation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil and cultivation: Outcome Check

Outcome check for Soil and cultivation: Explain how to prepare soil and use fertilizers properly, practice soil preparation and fertilizer use, and show respect and care for fertilizers to protect crops and the environment.

1. Name two soil components.
2. Compare two soil properties.
3. Explain one way to protect soil from damage.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Soil and cultivation: Outcome Check

Outcome check for Soil and cultivation: Identify steps of soil preparation for cultivation.

1. Name two soil components.
2. Compare two soil properties.
3. Explain one way to protect soil from damage.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants and environment: Lesson Opener

Learning focus: Plants and environment

Country link: a safe Rwandan observation, model, data table, or investigation for Plants and environment.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- name and explain uses of crops and trees, describe causes and effects of deforestation, practice caring for trees, and show willingness to protect and care for the environment
- explain the uses of trees
- discuss the uses of different types of crops
- discuss the causes and effects of deforestation
- explain ways to take care of trees
- recognize the types of crops

Inquiry questions:

- How can plants and environment help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants and environment: Learn

Plants and environment is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Plants and environment
- Plants
- environment
- concept
- observation
- evidence
- safety
- conclusion

Plants and environment: Worked Example

Investigation model for Plants and environment: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Plants and environment: Vocabulary And Study Skill

Use these words and study moves before you practise Plants and environment. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Plants and environment
- Plants
- environment
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Plants and environment without a clear example, method, or evidence.

Plants and environment: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Plants and environment.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Plants and environment: Practice And Correction

Practice for Plants and environment:

Main outcome: name and explain uses of crops and trees, describe causes and effects of deforestation, practice caring for trees, and show willingness to protect and care for the environment.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Plants and environment: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Plants and environment.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants and environment: Outcome Check

Outcome check for Plants and environment: Name and explain uses of crops and trees, describe causes and effects of deforestation, practice caring for trees, and show willingness to protect and care for the environment.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants and environment: Outcome Check

Outcome check for Plants and environment: Explain the uses of trees.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human digestive system: Lesson Opener

Learning focus: Human digestive system

Country link: a safe Rwandan observation, model, data table, or investigation for Human digestive system.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- identify the parts and functions of the digestive system, explain the stages of digestion, apply hygiene practices, prepare a balanced diet, and show positive habits to keep their digestive system healthy
- identify major parts of the digestive system
- explain the functions of the digestive system
- explain stages of digestion
- identify components and elements of balanced diet
- recognize parts of the digestive system and their functions

Inquiry questions:

- How can human digestive system help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human digestive system: Learn

Human digestive system is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Human digestive system
- Human
- digestive
- system
- concept
- observation
- evidence
- safety

Human digestive system: Worked Example

Worked example for Human digestive system: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Human digestive system: Vocabulary And Study Skill

Use these words and study moves before you practise Human digestive system. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Human digestive system
- Human
- digestive
- system
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human digestive system without a clear example, method, or evidence.

Human digestive system: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human digestive system.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human digestive system: Practice And Correction

Practice for Human digestive system:

Main outcome: identify the parts and functions of the digestive system, explain the stages of digestion, apply hygiene practices, prepare a balanced diet, and show positive habits to keep their digestive system healthy.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human digestive system: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human digestive system.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human digestive system: Outcome Check

Outcome check for Human digestive system: Identify the parts and functions of the digestive system, explain the stages of digestion, apply hygiene practices, prepare a balanced diet, and show positive habits to keep their digestive system healthy.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human digestive system: Outcome Check

Outcome check for Human digestive system: Identify major parts of the digestive system.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human diseases: Lesson Opener

Learning focus: Human diseases

Country link: a safe Rwandan observation, model, data table, or investigation for Human diseases.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- define different types of diseases, identify common diseases in the community, explain ways to prevent them, recognize symptoms, and show positive attitudes by practicing and promoting disease prevention
- define infectious, non-infectious and chronic diseases
- identify common infectious, non- infectious and chronic diseases in the community
- explain simple ways to prevent infectious, non- infectious and chronic diseases
- preventing different kinds of diseases by practicing good hygiene and habits
- noticing changes in the body or behavior that could indicate illness

Inquiry questions:

- How can human diseases help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human diseases: Learn

Human diseases is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Human diseases
- Human
- diseases
- concept
- observation
- evidence
- safety
- conclusion

Human diseases: Worked Example

Investigation model for Human diseases: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Human diseases: Vocabulary And Study Skill

Use these words and study moves before you practise Human diseases. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Human diseases
- Human
- diseases
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human diseases without a clear example, method, or evidence.

Human diseases: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human diseases.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human diseases: Practice And Correction

Practice for Human diseases:

Main outcome: define different types of diseases, identify common diseases in the community, explain ways to prevent them, recognize symptoms, and show positive attitudes by practicing and promoting disease prevention.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human diseases: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human diseases.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human diseases: Outcome Check

Outcome check for Human diseases: Define different types of diseases, identify common diseases in the community, explain ways to prevent them, recognize symptoms, and show positive attitudes by practicing and promoting disease prevention.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Human diseases: Outcome Check

Outcome check for Human diseases: Define infectious, non-infectious and chronic diseases.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electricity: Lesson Opener

Learning focus: Electricity

Country link: a safe Rwandan observation, model, data table, or investigation for Electricity.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain how electricity is made and why it is important, identify sources and parts of electric circuits, build and manage simple circuits, and show safe habits when working with electricity
- explain the importance and production of electricity
- identify different sources of electricity
- list common tools and materials used in electricity work
- identify components of electric circuit
- construct a simple electric circuit

Inquiry questions:

- How can electricity help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electricity: Learn

Electricity explains moving electric charge in circuits.

Core idea:

- A complete circuit is needed for current to flow.
- Cells, wires, switches, and bulbs can show current effects safely.
- Electricity can produce light, heat, sound, movement, and magnetism.
- Safety matters: never experiment with mains electricity.

Key words:

- Electricity
- concept
- observation
- evidence
- safety
- conclusion

Electricity: Worked Example

Circuit model for Electricity: Light a bulb safely.

1. Use a cell, two wires, a switch, and a small bulb.
2. Connect a complete circuit.
3. Close the switch and observe the bulb.
4. Open the switch and observe again.

Conclusion: current flows only when the circuit is complete.

Electricity: Vocabulary And Study Skill

Use these words and study moves before you practise Electricity. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electricity
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electricity without a clear example, method, or evidence.

Electricity: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electricity.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electricity: Practice And Correction

Practice for Electricity:

Main outcome: explain how electricity is made and why it is important, identify sources and parts of electric circuits, build and manage simple circuits, and show safe habits when working with electricity.

1. Draw a complete circuit with a cell, switch, and bulb.
2. Explain why a broken circuit stops current.
3. List two electrical safety rules.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electricity: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electricity.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electricity: Outcome Check

Outcome check for Electricity: Explain how electricity is made and why it is important, identify sources and parts of electric circuits, build and manage simple circuits, and show safe habits when working with electricity.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electricity: Outcome Check

Outcome check for Electricity: Explain the importance and production of electricity.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials and state of matters: Lesson Opener

Learning focus: Materials and state of matters

Country link: a safe Rwandan observation, model, data table, or investigation for Materials and state of matters.

Progression focus: add reasons, compare examples, and explain choices more independently; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- classify materials as metals or non-metals based on their properties, calculate their density, and demonstrate the changes of state of matter
- identify common metals
- classify common metals based on their properties
- state the uses of metals
- name objects made from metals
- describe and recognize the properties of metals and use them safely

Inquiry questions:

- How can materials and state of matters help you solve a real problem in Science and Elementary Technology?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials and state of matters: Learn

Materials and state of matters is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Materials and state of matters
- Materials
- state
- matters
- concept
- observation
- evidence
- safety

Materials and state of matters: Worked Example

Investigation model for Materials and state of matters: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Materials and state of matters: Vocabulary And Study Skill

Use these words and study moves before you practise Materials and state of matters. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Materials and state of matters
- Materials
- state
- matters
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Materials and state of matters without a clear example, method, or evidence.

Materials and state of matters: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Materials and state of matters.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Materials and state of matters: Practice And Correction

Practice for Materials and state of matters:

Main outcome: classify materials as metals or non-metals based on their properties, calculate their density, and demonstrate the changes of state of matter.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Materials and state of matters: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Materials and state of matters.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials and state of matters: Outcome Check

Outcome check for Materials and state of matters: Classify materials as metals or non-metals based on their properties, calculate their density, and demonstrate the changes of state of matter.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials and state of matters: Outcome Check

Outcome check for Materials and state of matters: Identify common metals.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Carpentry tools: Review Clinic 1

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Carpentry tools that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Masonry tools: Review Clinic 2

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Masonry tools that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Light: Review Clinic 3

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Light that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Computer memories and storages: Review Clinic 4

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Computer memories and storages that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Science and Elementary Technology.

- Carpentry tools: Explain this term using an example from the book, your school, or your community.
- Masonry tools: Explain this term using an example from the book, your school, or your community.
- Light: Explain this term using an example from the book, your school, or your community.
- Computer memories and storages: Explain this term using an example from the book, your school, or your community.
- Writing skills on computer: Explain this term using an example from the book, your school, or your community.
- Computer research: Explain this term using an example from the book, your school, or your community.
- Programming for children: Explain this term using an example from the book, your school, or your community.
- Water pollution: Explain this term using an example from the book, your school, or your community.
- Soil and cultivation: Explain this term using an example from the book, your school, or your community.
- Plants and environment: Explain this term using an example from the book, your school, or your community.
- Human digestive system: Explain this term using an example from the book, your school, or your community.
- Human diseases: Explain this term using an example from the book, your school, or your community.
- Electricity: Explain this term using an example from the book, your school, or your community.
- Materials and state of matters: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Science and Elementary Technology answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Science and Elementary Technology that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.